07 Beyond democratisation claims: AI, expertise and productive friction in AUTOMATA

Nevio Dubbini (nevio.dubbini@unipi.it), Gabriele Gattiglia

Claims that AI democratises archaeology risk obscuring how expertise and control are redistributed through automation. Free, Libre, and Open Source (FLOS) principles provide a basis for examining who can inspect, question, and modify the computational criteria by which archaeological data are produced and evaluated.

We examine this process through the lens of our work within AUTOMATA, a Horizon Europe project that develops AI- and Robotics-driven automated, enriched digitisation of archaeological ceramics and lithics. Its AI-based quality-assessment framework combines statistical indicators, machine learning, and explainability to assess 3D, hyperspectral, XRF, and Raman acquisitions and to support decisions on acceptance, adjustment, reacquisition, or expert review. Specialists formalise judgements through reference datasets, features, labels and acceptance thresholds. We argue that specialists should act as critical interlocutors, introducing productive friction by bringing complexity, material variability, contextual knowledge and interpretative uncertainty into dialogue with computational criteria and questioning how data quality is defined. The framework's provisions for reviewing classifications and updating criteria offer opportunities to develop this critical role.

Although all code developed within AUTOMATA will be released as open source and all project data will be openly available through the European Collaborative Cloud for Cultural Heritage (ECCCH), supported by ECHOES, openness alone does not establish a more equitable distribution of expertise and control. Making validation procedures, decision rules and data exclusions explicit is necessary for scrutiny but does not resolve inequalities in the capacity to challenge and change them. Without addressing who has the expertise, resources and authority to exercise these freedoms, appeals to openness risk legitimising existing distributions of control under the language of democratisation.